Computational HW 2

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Realistic Projectile Motion

I Introduction

Consider two identical objects that start at the same height above the ground. One object is dropped, and the other is launched horizontally.

Including air resistance, we quantify which object hits the ground first. This paper explores the dependence of landing time difference on the parameters mass, starting height, and launch velocity.

II Methods

Without drag, the equations of motion for a projectile can be modeled as four differential equations:

$$\frac{dx}{dt} = v, \qquad \frac{dv_x}{dt} = 0$$

$$\frac{dy}{dt} = v, \qquad \frac{dv_y}{dt} = -g$$

However, factoring in drag, we have two components to consider:

$$F_{D,y} = \frac{v_y}{v} F_D \rightarrow -v_y v B_2$$

$$F_{D,x} = \frac{v_x}{v} F_D \rightarrow -v_x v B_2$$

These forces then act and update the x and y velocities, giving rise to four equations of motion in Euler's method:

$$x_{i+1} = x_i + v_{x,i}\Delta t \tag{1}$$

$$y_{i+1} = y_i + v_{y,i}\Delta t \tag{2}$$

$$v_{x,i+1} = v_{x,i} - \frac{B_2}{m} v_i v_{x,i}\Delta t \tag{3}$$

$$v_{y,i+1} = v_{y,i} - \left(g + \frac{B_2}{m} v_i v_{y,i}\right)\Delta t \tag{4}$$

where $B_2 = (0.5)\ c\rho A$.

The parameters c, ρ, A correspond to the drag coefficient, air density, and cross-sectional area.

With these, there is no analytic solution to the equations of motion, and so numerical methods must be used.

The standard Euler's method is used, where the four first-order differential equations are iterated through as shown in Algorithm 1.

Algorithm 1 Euler's Method

Require: $y > 0$
Ensure: time array t
  $t \leftarrow [\ ]$ ▷ Initialize empty list
  $(x, y, v_x, v_y, v) \leftarrow$ Initial Values
  $i \leftarrow 0$
  repeat
    $x_{i+1} = x_i + v_{x,i}\Delta t$
    $y_{i+1} = y_i + v_{y,i}\Delta t$
    $v_{x,i+1} = v_{x,i} - B_2 v_i v_{x,i}\Delta t/m$
    $v_{y,i+1} = v_{y,i} - (g + B_2 v_i v_{y,i})\,\Delta t/m$
    $t_{i+1} \leftarrow t_i + \Delta t$
    $i \leftarrow i + 1$
  until $y_i < 0$
  return t

We want to quantify the time difference between the two objects hitting the ground. Our numerical method cannot find the value when height exactly equals zero, so we must interpolate between the last and second-to-last points. For the shot object, we interpolate between the final two x-positions.

Algorithm 2 Shot Landing Time Interpolation

Require: $y_{n-1} > 0, y_n < 0$
Ensure: Landing Time t_L
  $t_L \leftarrow [\ ]$ ▷ Initialize empty list
  $x_L \leftarrow \dfrac{x_n - x_{n-1}}{|y_n + y_{n-1}|} y_{n-1} + x_{n-1}$
  $t_L \leftarrow t_{n-1} + \dfrac{x_L - x_{n-1}}{v_{x,n-2}}$
  return t_L

For the dropped object, we extrapolate from the latest positive height to the ground.

$$t_{L,drop} = t_{n-1} + |\frac{y_{n-1}}{v_{y,n-1}}|$$

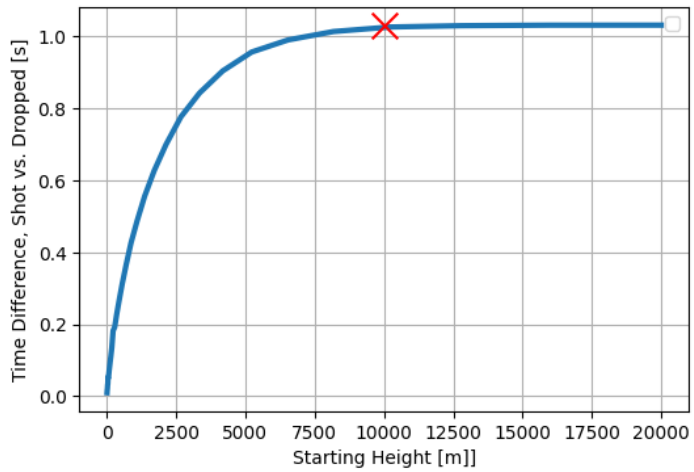
From here, we explore the dependence of the difference in landing times $\Delta t_L = t_{L,shot} - t_{L,drop}$ on the parameters $m, y_0, v_{x,0}$, by generating initial values ranging from 10 to 10000 for each control parameter.

The baseline object was a sphere:

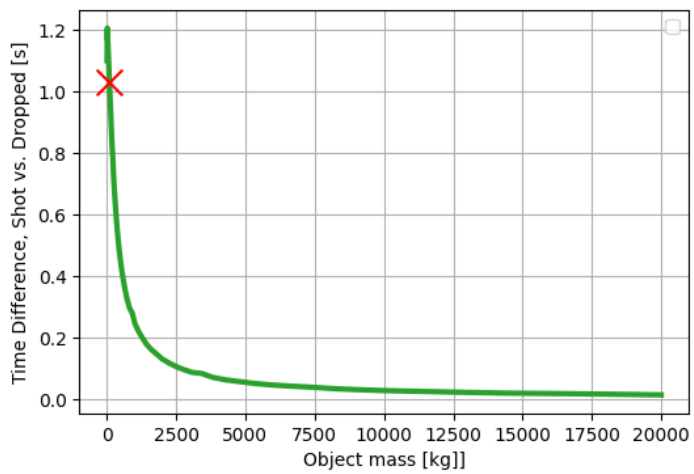
1. $m = 100$ kg
2. $y_0 = 10000$ m
3. $v_{x,0} = 100$ m/s (0 for dropped object)
4. $A = 0.1$ m^2
5. $\rho = 1.225$ kg/m^3
6. $c = 0.5$

No generative AI was used in this project.

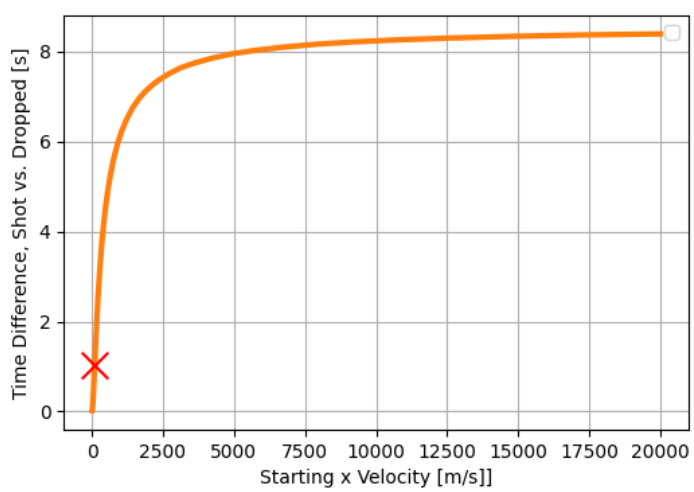
III Results



(a) Dependence on starting height



(b) Dependence on object mass



(c) Dependence on initial x velocity

Figure 1: Control parameter vs. Landing Time Difference for different control parameters y_0, m, $v_{x,0}$, ranging from values 10-20000. Red "$\times$" marker indicates the baseline objects, as discussed in Section II.

IV Discussion

All control parameters were found to have a dwindling effect on landing time difference in the extreme limits ($\sim$10,000). This agrees with the numerical model shown in Section II. Shown in each figure as a red "$\times$" marker, a baseline test is shown, where $\Delta t_L \approx 1.03$.

As the drag coefficient goes to zero, both objects land at the same time. This causes B_2 to equal zero. This is shown in both equation (2) and (4), with

$$\lim_{c \to 0} \frac{B_2}{m} v_i v_{x,i} = 0$$

This removes the dependence on velocity v in both equations of motion, thus uncoupling the x and y motion, returning to a simplified projectile motion.

i Initial Height y_0

With no ground, both objects have identical paths of motion as their initial height changes. As time passes, the shot object reaches the ground slightly later than the dropped object.

At small height, the objects reach the ground at nearly the same time, but we do not see any interesting effects or consequences from the landing time depending on the height. However, increasing the starting height clearly forces the objects to spend longer in free fall - therefore slowly increasing the separation.

Eventually, the objects stop accelerating. At large heights, a small change in height barely changes the Landing Time Difference (Δt_L), as the extra height doesn't allow for a (proportionally) large space to accelerate.

Furthermore, The objects reach terminal velocity, where their y-velocity does not increase any further. This is shown in Equation (4), where the term $g + \frac{B_2}{m} v_i v_{y,i}$ goes to zero, and both objects approach zero acceleration.

ii Object Mass m

As we know from basic kinematics, heavy objects are acted on by gravity harder, and thus they accelerate faster. Accounting for drag, we see similar results.

As the masses of two objects increase, their terminal velocity increases. Furthermore, drag affects the motion less, as shown by a classic $1/m$ relationship.

With larger mass, the term $\frac{B_2}{m} v_i v_{y,i}$ decreases, so the object can reach larger speeds before reaching terminal velocity. In addition, smaller masses are significantly affected by drag.

Again by the inverse relationship, lighter masses quickly reach terminal velocity. Due to the shot object with $v_x \neq 0$, this terminal y velocity is smaller, and thus falls slower.

iii Horizontal Velocity $v_{x,0}$

Perhaps the most interesting result is the dependence on initial horizontal velocity. First, as expected, at near-zero speeds, the shot object perfectly mirrors the dropped, and Δt_L goes to zero. The dropped object will always land first, unless $v_{x,0} = 0$.

In the baseline case, the objects are dropped from 10,000 meters, and a velocity of 100 m/s causes a difference in time of 1 s, after a free-fall time of ~ 68.5.

Though minor, there is a clear and measurable difference in landing time if one object is given an initial horizontal velocity.

As this initial horizontal velocity increases, Δt_L increases, but plateaus and changes slowly.

Terminal y velocity is reached when the term in eq. (4) changing with time is zero, meaning

$$g = \frac{B_2}{m} v v_y$$

A large x velocity in turn creates a large speed v. So, the y terminal velocity is (comparatively) slow.

Conversely, in the $v_x = 0$ case, there is no x component contributing to the speed v, so the y velocity can reach a much higher value before zero net force acts on the object.

As the initial x velocity increases, there is less and less of an effect on landing time due to the inverse relationship between v and v_y. Initially, there is a great effect on Δt_L, but plateaus as $v_{x,0} \to 0$.

V Conclusion

Through the use of the Euler method, we successfully model the difference in landing times between a launched object versus a dropped object.

From there, we analyze and conceptually reason the dependence on parameters of starting height, object mass, and initial horizontal velocity. It was found that, due to drag, a launched object lands later than a dropped object.

The largest contributor to the discrepancy was found to be $v_{x,0}$ due to the impact on terminal y velocity, as well as extremely small object masses.

Conversely, large orders of magnitudes for each control parameter resulted in minor changes in Δt_L.